Real-Time Hand Gesture Recognition by Decoding Motor Unit Discharges Across Multiple Motor Tasks From Surface Electromyography

Chen Chen, *Member, IEEE*, Yang Yu, Xinjun Sheng, *Member, IEEE*, Jianjun Meng, and Xiangyang Zhu

Abstract*—*Objective. **Surface electromyography (EMG) decomposition techniques have been developed to decode motor neuron activities non-invasively in the past decades, showing superior performance in human-machine interfaces such as gesture recognition and proportional control. However, neural decoding across multiple motor tasks and in real-time remains challenging, which limits its wide application. In this work, we proposed a real-time hand gesture recognition method by decoding motor unit (MU) discharges across multiple motor tasks (>10) in a motion-wise way.** ***Methods.*** **The EMG signals were first divided into numerous segments related to motions. The convolution kernel compensation algorithm was applied for each segment individually. The local MU filters, which indicate the MU-EMG correlation for each motion, were calculated iteratively in each segment and reused for global EMG decomposition to trace the MU discharges across motor tasks in real-time. The motion-wise decomposition method was applied on the high-density EMG signals recorded during twelve hand gesture tasks from eleven non-disabled participants. The neural feature of discharge count was extracted for gesture recognition based on five common classifiers.** ***Main results.*** **On average, 164±34 MUs were identified for twelve motions from each subject, with a pulse-to-noise ratio of 32.1±5.6 dB. The average time cost of EMG decomposition in a sliding window of 50 ms was less than 5 ms. The average classification accuracy using a linear discriminant analysis classifier was 94.6±8.1%, which was significantly higher than that of a time-domain feature called root mean square. The superiority of the proposed method was also validated with a previously published EMG database comprising 65 gestures.** ***Conclusion and Significance.*** **These results indicate the feasibility and superiority of the proposed method for MU identification and hand gesture recognition across multiple motor tasks, extending the potential applications of neural decoding in human-machine interfaces.**

***Index Terms*—Gesture recognition, surface electromyography, motion-wise decomposition, motor unit, human-machine interface.**

Manuscript received 16 February 2022; revised 1 September 2022 and 11 December 2022; accepted 3 January 2023. Date of publication 6 January 2023; date of current version 20 June 2023. This work was supported in part by the China National Key R&D Program under Grant 2020YFC207800, in part by the National Natural Science Foundation of China under Grants 91948302 and 52205024, and in part by the China Postdoctoral Science Foundation under Grants BX2021180 and 2022M722103. *(Corresponding author: Xiangyang Zhu.)*

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This article has supplementary downloadable material available at https://doi.org/10.1109/TBME.2023.3234642, provided by the authors.

Digital Object Identifier 10.1109/TBME.2023.3234642

I. Introduction

SURFACE electromyography (EMG) has found wide applications in human-machine interfaces such as prosthetics, exoskeletons, and virtual reality systems [1], [2], [3], [4]. The EMG signals derive from the motor neuron discharges in the spinal cord and, therefore, reflect the neural drive into the muscles [5]. Different muscles are activated during human movements, exhibiting the multiple patterns in the EMG signals [6]. Those patterns could be recognized by machine learning techniques, where various classifiers are applied after extracting EMG features. The time- or frequency-domain features are usually used [7], [8]. However, those features could only interpret partial neural activities since they are extracted from the EMG amplitudes or frequency spectrums. The amplitude cancellation and crosstalk would result in the information loss of neural drive [9], [10].

In the past two decades, decomposition techniques have been developed to decode the neural activities underlying the surface EMG signals in a non-invasive way [11], as shown in Fig. 1. The motor unit (MU) discharges, which directly represent the neural commands sent by the central nervous system in binary codes [12], [13], are usually identified from the surface EMG signals based on blind source separation or template matching methods [14], [15], [16]. The decoded MU action potential trains (MUAPTs) have been validated in various contraction levels and muscles, showing promising performance in human-machine interaction systems [17], [18], [19]. Different MUs can be projected to multiple motions. At the same time, the discharge rate provides the activation level of the corresponding motion. The MUAPT has been used to predict sequential hand gestures and continuous movements of finger, wrist, ankle, and

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knee [17], [19], [20], [21], [22]. Higher classification accuracy and lower estimation error were obtained with MUAPTs in previous studies.

However, there are still several limitations of the decomposition-based interfaces. Current decomposition methods could only decode a fraction of activated MUs. The number of output MUs in each decomposition is limited even though more movements are involved. Accordingly, the current investigations mainly focused on a few motions or degree-of-freedoms (DoF), where the wrist and finger movements were usually investigated separately [20], [22], [23]. Furthermore, the decomposition results exhibit some uncertainty in the order and amplitude of sources due to the inherent property of blind source separation. Additional clustering methods are usually needed to unify the feature dimension extracted from MU activities [22]. In addition, it is difficult to trace the MU discharges across different datasets, especially when multiple motions are involved. The multi-channel MU action potentials (MUAPs) are usually extracted after each decomposition to track the MU discharges across two tasks [24], [25] and group the MU discharges based on the spatial distribution of the MUAPs [17].

The real-time analysis capacity is another issue limiting the broad application of EMG decomposition in human-machine interfaces. The classification performance for human movements based on the EMG decomposition has been mainly demonstrated offline [17], [26]. Our previous work evaluated the MU activities during discrete motor tasks and proposed an MU-classification method for gesture recognition [26]. The decoded MUs were associated with twelve motions by classifying the MUAPs. The high classification accuracy of the proposed method demonstrated the efficiency and superiority of EMG decomposition in gesture recognition. However, the presented results were limited to an offline analysis due to the computation complexity in the EMG decomposition and the MUAP extraction. Attempts have been made to fulfill the real-time MU decoding recently [21], [27], [28]. Specifically, the separation matrix of MUs is calculated in an offline initialization procedure. Then the MU discharges are extracted directly using the obtained separation matrix in the real-time implementation. To date, the real-time decomposition schemes have been validated during simulated conditions or simple movements. It remains unknown whether the decomposition technique can decompose EMG signals of multiple motions in real-time.

In this study, we applied the MU filters, which were calculated in one motor task, to identifying MU discharges across multiple tasks. The so-called motion-wise decomposition could trace the MU discharges across multiple motions and is suitable for real-time applications. Then we proposed a gesture recognition scheme and demonstrated its superiority in the discrimination of multiple hand gestures.

II. Methods

A. Experiments

1) Participants: Eleven able-bodied subjects (eight males and three females, aged 24.1±2.4 years) participated in the experiments without neurological or psychiatric disorders. The experimental protocol followed the Declaration of Helsinki and was approved by the local ethic committee of Shanghai Jiao Tong University (approval numbers B2020026I).

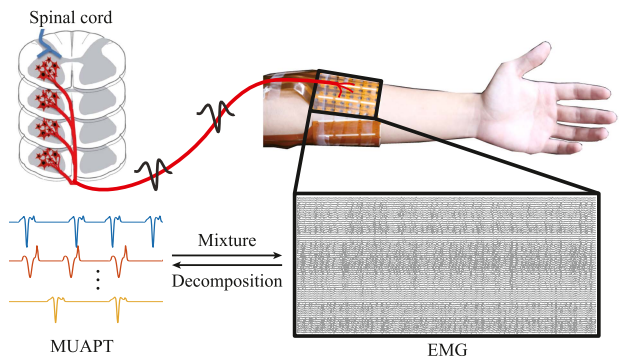

Fig. 1. Illustration of the MUAPTs derived from the spinal cord.

2) Acquisition Setup: High-density surface EMG signals were recorded from the forearm muscles [26]. The structure of the electrodes contains three grids of 64 electrodes (8×8 with an inter-electrode distance of 10 mm in two directions). The diameter of each electrode is 3 mm. The electrode grids were mounted around the proximal third of the forearm of the dominant arm, covering nearly the entire forearm circumference (Fig. 1). For three subjects with small forearm circumference, one or two columns of the grids were discarded from the analysis. Prior to the electrode placement, the skin of the examined regions was cleaned using alcohol pads. The flexible surface electrode array was attached to the forearm with a compatible, double-sided adhesive sticker after gel preparation. The electrode array was further secured with a medical tape. Surface EMG signals were amplified and sampled at 2048 Hz with a gain of 500 and a 12-bit resolution (EMG-USB2+, OT Bioelettronica, Italy).

3) Experimental Protocol: The subjects were instructed to perform twelve motor tasks (including rest) with their dominant hands, as shown in Fig. 2. The first eleven motor tasks included hand close, hand open, wrist flexion, wrist extension, radius deviation, ulnar deviation, forearm pronation, forearm supination, hand pinch with two fingers, hand pinch with three fingers, and key grip (Motion 1-11). The rest is indicated with Motion 12. The subjects were informed to maintain the medium contraction force without any resistance for each movement.

During the experiment, the subjects were seated in a chair with the arm fully extending and pointing toward the ground and the palm facing inward. There were eight trials for each subject. In each trial, subjects were instructed to perform each motion and maintain the gesture sequentially. Each motor task was sustained for 10 s with a 5-s rest between tasks. The picture of the motions was displayed to instruct the subject. The order of the first eleven tasks was randomized for each participant, with the rest task performed at last.

Before the experiment, each subject had sufficient time to familiarize themselves with the experimental procedure and the motor tasks. During the experiment, participants were seated comfortably in a height-adjustable chair in an upright position

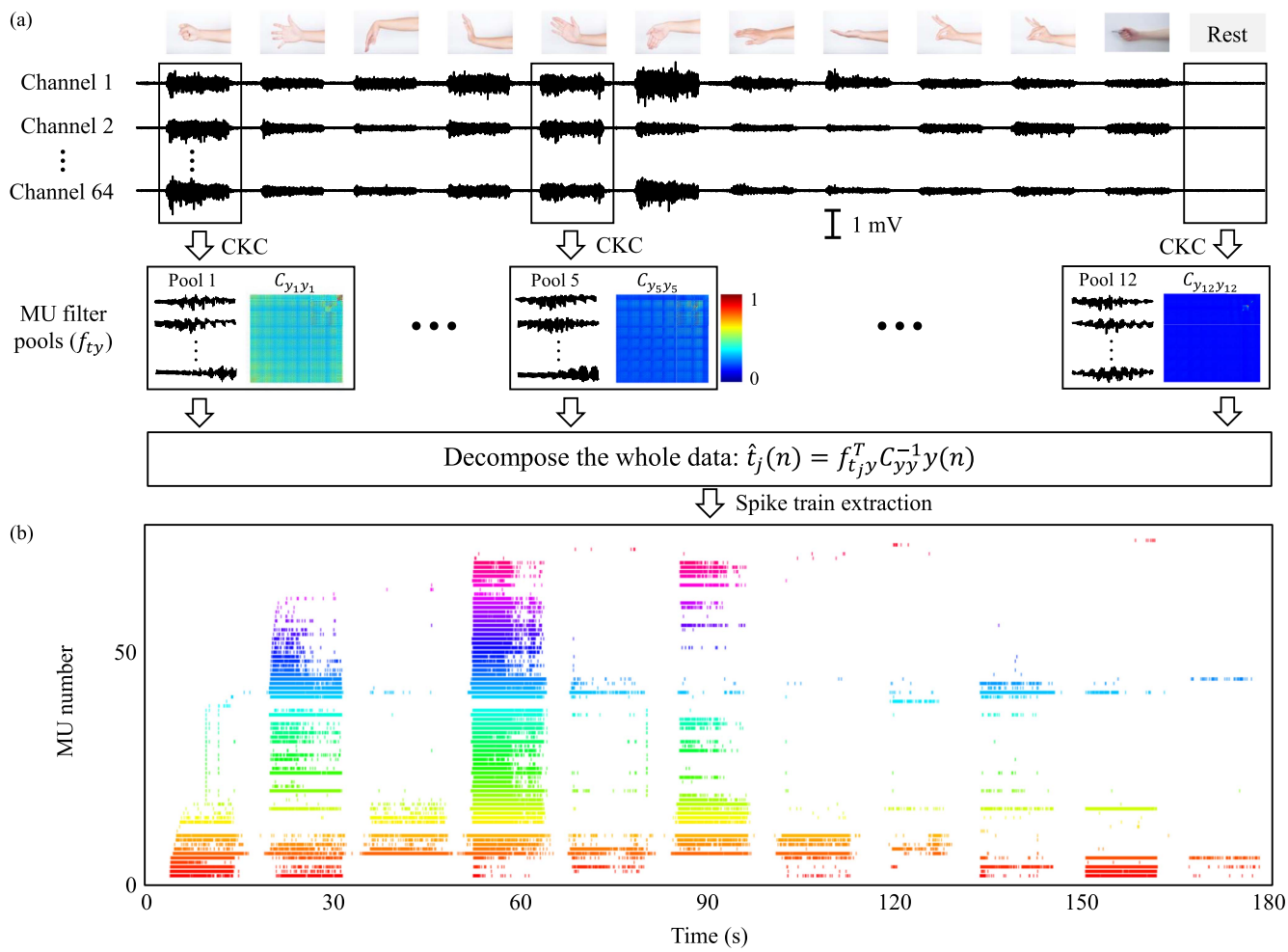

Fig. 2. Example results of motion-wise decomposition for each electrode grid. (a) The twelve motor tasks and the corresponding EMG signals of a grid. The EMG signals were divided into twelve segments to calculate the MU filters. Then the MU filters were grouped to decompose the global data. (b) The identified MUAPTs decoded from the global data. Each vertical bar depicts one motor unit discharge, with different MUAPTs shown in different colors. The MUAPTs are in alignment with the EMG signals on the horizontal axis.

with both of their upper limbs resting and pointing towards the ground. Each trial lasted 3 min, with a rest of 1-2 minutes between trials.

B. EMG Decomposition

The EMG signals were decomposed into MUAPTs with a motion-wise decomposition algorithm (Fig. 2) [29]. Briefly, the EMG signals from one trial were firstly decomposed in an offline manner, with the MU filters collected and stored. Then the MU filters were applied to all eight trials to estimate MUAPTs in real-time. Fig. 3 shows the EMG decomposition and classification scheme. The detail of the motion-wise decomposition algorithm is introduced in the Appendix.

1) Offline Decomposition: The EMG signals were band-pass filtered between 20 and 500 Hz using a 4th-order Butterworth filter in the preprocessing step. A comb filter was applied to remove the power line interference of 50 Hz. The EMG channels with abnormal amplitudes caused by poor contact between electrode and skin were selected and discarded automatically based on their root mean square values.

The motion-wise decomposition was implemented for each electrode grid individually, and then the decomposition results were combined. A few duplicated MUAPTs (usually <2) might be identified from two adjacent grids. We did not remove these duplications since the proportion of duplications was insignificant compared with the quantity of all MUs. The preprocessed EMG signals were divided into twelve segments of 15-s signals. Then the EMG signals were decomposed into MUAPTs using the algorithm described in Appendix. The MUAPT were filtrated depending on their discharge properties. The MUAPTs whose average discharge rate did not lie between 5 and 50 Hz were discarded. The MUAPTs with the pulse-to-noise ratio (PNR) <25 dB and coefficient of variation of inter-spike interval (CoV_{ISI}) >0.3 were also removed from the following analysis. The average percentage of MUAPT removed based on discharges and PNR was 58.4±8.4%, compared with the number of remaining MUAPT after duplication removal. The decomposition parameters of the remaining MUAPTs were kept for the online decomposition.

2) Online Decomposition: One out of eight trials was selected for the offline decomposition to obtain the decomposition

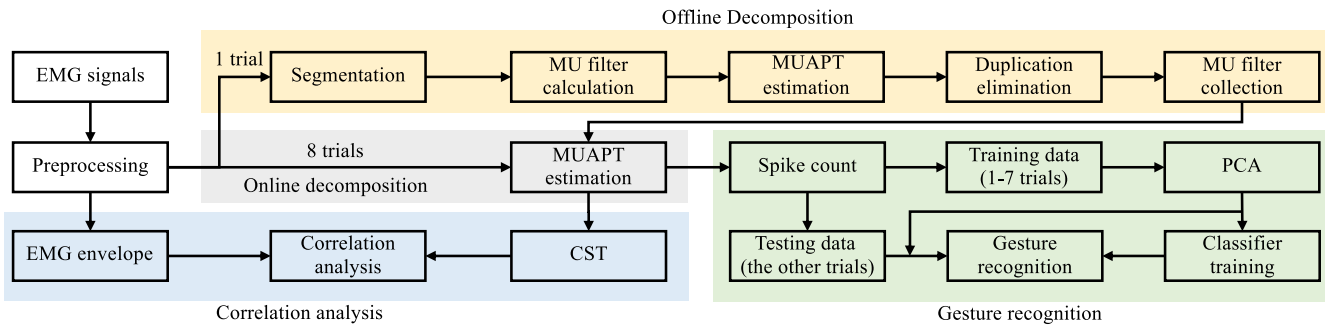

Fig. 3. The flow chart of motion-wise decomposition for classification. For each subject, the EMG signals from all eight trials were first preprocessed. One trial was selected randomly for offline decomposition to obtain the MU filters. Then the MU filters were applied for the online decomposition of the EMG signals from all remaining trials. The neural features (spike count) were extracted from eight trials and then divided into training and testing datasets. The features selected randomly from n out of 8 trials ($n = 1, 2, \ldots, 7$) were used to train the classifier after PCA. The features from the remaining trials were used to test the classification performance. The envelope and cumulative discharge rate were extracted from EMG and MUAPTs respectively for correlation analysis.

parameters. Then the EMG signals from the remaining trials were decomposed online with a sliding window of 50 ms. There was no overlapping between adjacent windows. The MUAPTs were estimated from each electrode grid depending on (2), where the MU filters ($f_{t_j y}$) and the correlation matrix (C_{yy}) were obtained from each grid in the offline trial. The discharges of each MU were extracted with the pre-trained clustering centroids. The EMG signals were no longer segmented, and the online decomposition was directly implemented on the global EMG covering the 12 motions. Each trial was selected for the offline decomposition once.

The classic CKC was also applied for comparison with the motion-wise decomposition method. The 180-s EMG signals from each trial were decomposed into MUAPTs without segmentation. The decomposition parameters and the offline/online configuration of classic CKC were the same as the motion-wise CKC (mwCKC). The decomposition results were compared between mwCKC and CKC, and we used three metrics, rate of agreement (RoA), sensitivity, and precision, to evaluate the common MUs [27], [30].

3) MUAP Extraction: The MUAP waveforms of each MU were extracted for each channel using the spike-triggered averaging (STA) method [31]. An interval of 128 samples centered by each MU discharge was extracted and averaged to estimate the waveform. The multi-channel MUAPs were weighed-averaged depending on the discharge number to measure the activation zone of each motion.

C. Data Analysis

The neural features extracted from the online decomposition results by spike counting were divided into two parts to train the classifier and test the classification performance. The correlation analysis was implemented to characterize the relationship between the neural features and the motion activation. The flow chart of data analysis is shown in Fig. 3.

1) Feature Extraction: The features were extracted from online decomposition results by counting the number of discharges of each MU in a sliding window (Fig. 2). The sliding window length was set as 200 ms with an overlapping of 150 ms between the adjacent windows. Only the central 8 s data in each 10 s motion were used to exclude the transient states of the motion. The dimension of the feature vector equaled the number of MUs. The feature space was reduced in dimensionality using the principal component analysis (PCA). The number of principal components used for the classification corresponded to $>99\%$ of explained variance. The average number of principal components kept for the classification was 76 ± 22, which was fixed during the online test.

2) Classification: Hand gesture recognition was evaluated on the linear discriminant analysis (LDA) classifier. The classifiers' input was the MUAPT features extracted from the online decomposition results. The MUAPT features were extracted from 8 trials for each subject. Random k trials ($1 \leq k \leq 7$) were included in the training set, while the remaining trials were used to test the classifier performance. The trial for offline decomposition was excluded from the testing data since the decomposition results in this trial had already involved the prior information of motion class. The eight trials, which were online-decomposed using the decomposition parameters from the same offline trial, constituted an individual dataset. The online decomposition results were obtained from different offline trials, resulting in 8 datasets for each subject. All the possible training and testing data combinations were implemented in each dataset. The classification accuracy was evaluated as the ratio between the number of correctly recognized samples and tested samples. Apart from the LDA, four other classifiers, support vector machine (SVM), K-nearest neighbor (KNN), bilayered neural network (BNN), and subspace discriminant (SD), were also applied for the gesture recognition. The classification results of these classifiers can be found in the supplementary materials.

For the comparison with the proposed method, the neural feature was also extracted from the decomposition results of the CKC. In addition, a time-domain EMG feature, root mean square (RMS), was extracted from each channel. The configuration of the PCA, sliding window, and the training/testing data was the same as that in the proposed method.

3) Correlation Analysis: The MUAPTs were combined to form the cumulative spike train (CST) [32]. The correlation between the MU discharge rate and the motion activation intensity

was analyzed. The MUAPTs of each motion were combined to form the CST [32], indicating the activation level of each MU population. The activation intensity of each motion was indicated by the EMG envelope, which was obtained by averaging and lowpass-filtering the rectified EMG signals across all channels. The cutoff frequency was set as 2 Hz. The discharge rate and the EMG envelope were normalized to [0,1] depending on their maximum values in each trial. The Pearson correlation coefficient and the root mean square error (RMSE) were calculated between normalized discharge rates and the EMG envelope.

4) Validation on EMG Database: To further validate the proposed gesture recognition scheme, we analyzed the classification performance on a high-density surface EMG database, consisting of 65 isometric hand gestures [33]. The EMG signals were recorded using 128 channels with two electrode grids positioned on the forearm of 20 non-disabled volunteers. The subjects performed 65 different hand gestures sequentially, including single finger or wrist movement and their combinations. The EMG signals were recorded in a differential manner, where consecutive channels were subtracted. The recording details of the EMG database could be found in [33].

We selected the first eleven subjects for validation. Due to the differential recording manner, the channels that are a multiple of 8 (channel 8, 16...) were removed, resulting in 112 channels for the following analysis. We divided the EMG signals into five trials as each hand gesture was repeated five times. Each trial lasted for 650 s, with 10 s for each gesture. The EMG signals in each trial were decomposed into MUAPTs with a segment length of 10 s. After decomposition, the contraction periods of each gesture were determined depending on the adjusted labels [33]. The other configuration of decomposition and feature extraction were the same as previously described. Four out of five trials were used to train the LDA classifier, while the remaining trial was used for testing. The RMS feature was also extracted for comparison.

All the decomposition and analysis were implemented in MATLAB 2021b (Matlab Inc. USA).

III. Results

A. Decomposition Results

The EMG signals recorded from multiple motions were decomposed into MUAPTs with the motion-wise decomposition scheme. On average, 164±34 MUs were identified during twelve motor tasks from each subject. The number of the identified MUs from three grids was 69±18, 57±16, and 38±17, respectively. The average PNR was 32.1±5.6 dB, which corresponds to an estimated decomposition sensitivity of about 90% and a false alarm rate of 2% [30]. More MUs with lower MUAP amplitudes were decoded by the mwCKC compared with the CKC, as shown in Fig. 4. On average, 10±5 MUAPTs were matched between the mwCKC and CKC, with an average RoA of 43.4±13.4%. The average sensitivity and precision were 72.0±16.8% and 54.9±17.2%, respectively. The average time cost of EMG decomposition in the sliding window was less than 5 ms, and the overall time cost for each trial (180 s) was less than 50 s. We provided two videos in the supplementary materials to demonstrate the online performance.

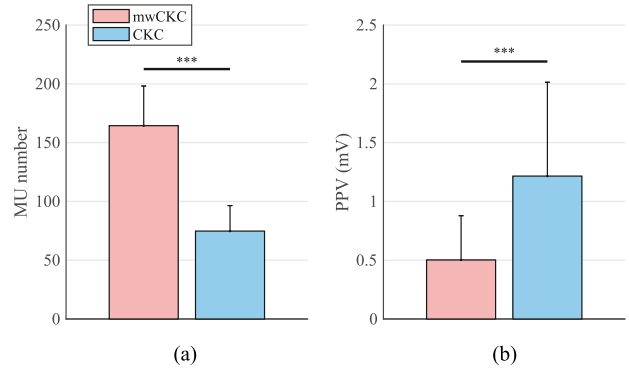

Fig. 4. The comparison of decomposition results between mwCKC and CKC. A and B illustrate the average MU number and peak-to-peak value (PPV) of MUAPs, respectively. A one-way analysis of variance was applied to evaluate the effect of the decomposition method on the MU number and PPV. The symbol $***$ indicates the significant difference with a level of ($P < 0.001$).

The detailed decomposition results about activated MUs in each motion are given in Table I. Each identified MU was activated during one or more motor tasks. Depending on the online decomposition results, there were 31±26 MUs activated during each motion. Among the twelve motions, most MUs were related to Motion 6 (Ulnar deviation), whereas the least MUs were activated during Motion 12 (rest). Over ten exclusive MUs were identified during two motions (wrist flexion and ulnar deviation). The weighted-averaging MUAP of each motion for a particular subject is shown in Fig. 5.

B. Classification Results

Fig. 6 illustrates the distribution of the first three principal components in the feature space from a particular subject. The classification accuracy for each motion is shown in Fig. 7. The average classification accuracy for each motion was 94.6±8.1% when using seven trials for the training. The recognition accuracy for Motion 12 (rest) was always the highest among the twelve motions for all classifiers. The classification accuracy based on the motion-wise decomposition method was significantly higher than that of the classic CKC method (47.3±38.0%).

The effect of the number of motions and training trials on the classification accuracy is shown in Fig. 9. Generally, the average classification accuracy decreased when more motions were involved, or less training data was used. The average classification accuracy for each motion was 89.5±10.4% (LDA) when using one trial for the training. The confusion matrix of classification accuracy using three methods (mwCKC, CKC, and RMS) is shown in Fig. 8.

C. Correlation Analysis

Fig. 10 demonstrates an example classification result in a trial. The correlation analysis results are shown in Fig. 11. The average correlation coefficient between the CST discharge rate and the

TABLE I
SUMMARY OF DECOMPOSITION RESULTS ABOUT ACTIVATED MUS IN EACH MOTION

	Offline		Online							
	No. MU (All)[1]	PNR (dB)	No. MU (All)	Grid 1	Grid 2	Grid 3	Exclusive[2]	PNR (dB)	Average DR[3]	PPV (mV)
Motion 1	30±17	29.9±4.6	31±18	10±11	8±6	17±14	6±5	28.0±4.0	18.5±14.9	0.281±0.182
Motion 2	38±18	29.1±3.4	39±18	2±6	17±12	16±10	5±6	27.5±3.1	15.9±12.5	0.298±0.178
Motion 3	59±30	30.9±4.3	60±31	20±20	9±9	32±19	19±15	28.9±3.6	26.6±15.9	0.351±0.242
Motion 4	50±24	31.4±4.9	52±24	19±13	36±24	10±12	6±6	29.0±3.8	24.0±16.0	0.418±0.258
Motion 5	24±15	28.8±3.5	24±13	3±6	9±7	4±6	11±10	27.0±3.4	16.4±13.3	0.291±0.230
Motion 6	61±32	30.7±4.1	61±31	11±12	25±15	29±18	7±6	28.7±3.5	23.8±15.9	0.366±0.224
Motion 7	26±15	29.2±4.1	27±17	4±7	12±13	6±7	9±8	27.5±3.7	17.3±13.7	0.329±0.246
Motion 8	16±9	29.3±4.2	17±11	3±5	6±7	6±6	4±4	27.4±4.1	13.7±11.8	0.299±0.243
Motion 9	16±9	28.0±3.2	17±11	2±5	10±8	4±6	3±4	26.5±3.5	14.4±11.8	0.283±0.186
Motion 10	19±12	28.3±3.7	20±15	3±7	11±7	5±10	5±5	26.7±3.5	18.3±14.2	0.288±0.200
Motion 11	17±16	28.5±3.9	18±17	2±7	8±9	6±9	3±4	26.7±3.6	18.7±15.0	0.279±0.199
Motion 12	3±2	28.3±3.0	3±2	0±0	0±1	1±2	2±2	24.2±4.7	13.4±12.3	0.086±0.089
Average	31±25	29.9±4.3	31±26	7±12	13±15	12±15	7±8	28.0±3.7	20.5±15.3	0.332±0.229

[1]: Each MU was regarded as activated if there were at least 20 discharges in the period of this motion.
[2]: Exclusive MU refers to that over 70% of discharges occurred in the period of only one motion.
[3]: DR: discharge rate.

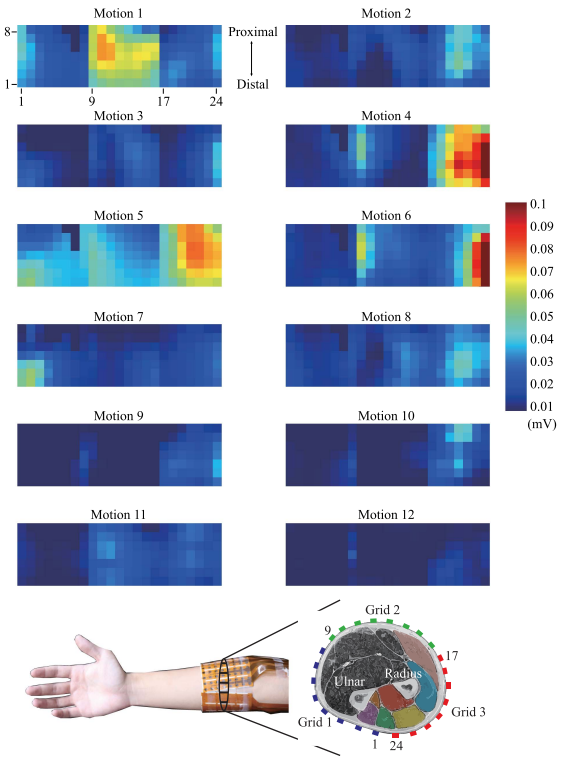

Fig. 5. The heatmap of discharge-weighted average MUAP from subject 11. The color indicates the peak-to-peak value of MUAP waveforms in 192 channels (8 rows × 24 columns). The spatial distribution of electrodes on the forearm circumference is shown in the bottom panel.

EMG envelope was higher than 0.8 for all motions, with the average RMSE lower than 0.2.

D. Database Results

On average, 313±213 MUs were identified during 65 motor tasks from each subject, with a PNR value of 28.0±8.9 dB. The number of the identified MUs from two grids was 110±43 and 203±182, respectively. The classification results of the EMG database are shown in Fig. 12. The average classification accuracy of 65 motions was 86.2±23.4% using the motion-wise

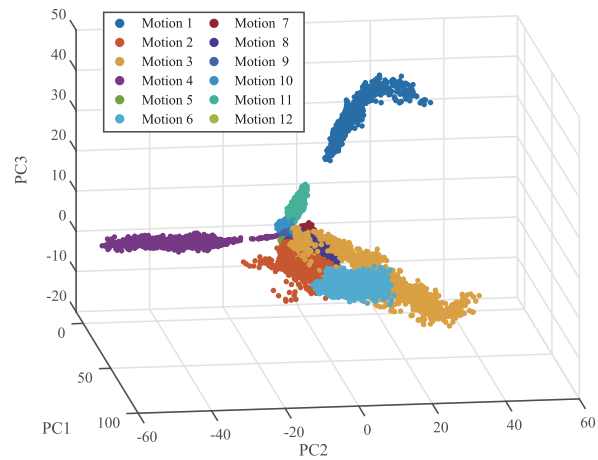

Fig. 6. The distribution of the neural feature from 8 trials of subject 11. Only the first three principal components are shown. Each point was calculated in a sample window. Each class is demonstrated with different colors. PC1 to PC3 indicate the first three principal components.

decomposition method, which was significantly higher than that of RMS feature (83.0±26.7%).

IV. DISCUSSION

In this work, we proposed a real-time pattern recognition scheme based on the EMG decomposition. The presented results showed that the motion-wise decomposition could be generalized to real-time conditions. The high classification accuracy proves the potential of the MU-based approach for gesture recognition, promoting the application of neural decoding techniques in human-machine interfaces.

A. EMG Decomposition for Multiple Motor Tasks

Previous studies have applied the EMG decomposition for neural decoding of various muscles and movements. However, most studies focused on only single-DoF tasks, where the same muscle groups were activated [24], [34]. One reason is that it is difficult to trace the MU discharges across motor tasks and trials. Due to the uncertainty of blind source separation and the random fluctuation in the discharge pattern, it is challenging to match the

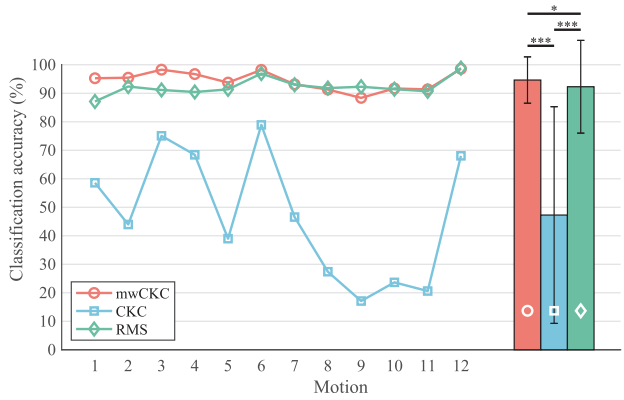

Fig. 7. The hand gesture recognition results of three features (mwCKC, CKC, and EMG feature). The lines in the left panel depict the classification accuracy of each motion, while the bars give the average results. The symbol $**$ and $***$ indicates the significant difference with a level of $(0.001 < P < 0.05)$ and $(P < 0.001)$, respectively.

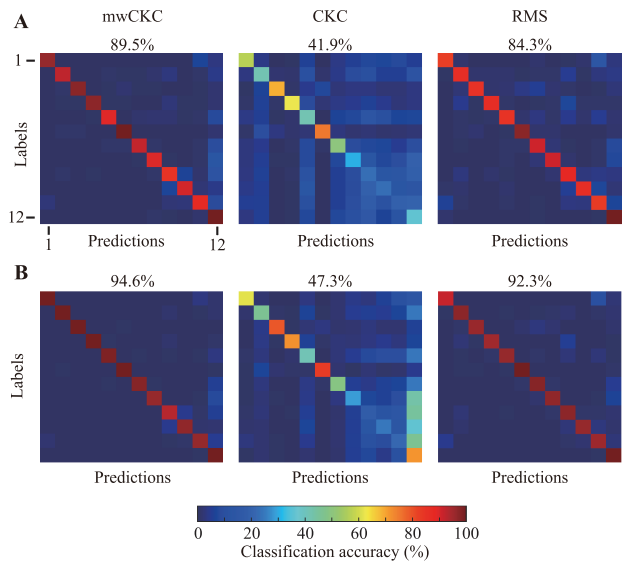

Fig. 8. The confusion matrix of classification accuracy of three methods averaged across all subjects. A and B show the classification results obtained with one and seven training trials, respectively. The color of each block in the confusion matrix shows the ratio of the number of predicted class (in x-axis) to the true class (in y-axis).

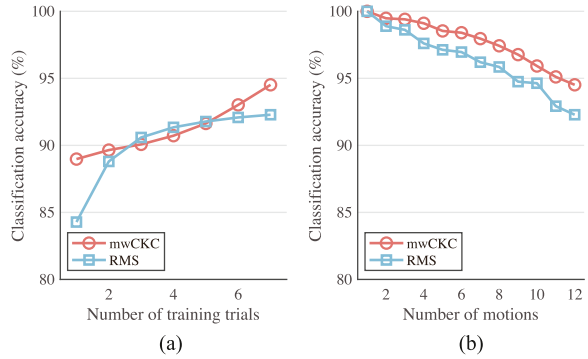

Fig. 9. The effect of the training trials (a) and number of motions (b) on the classification accuracy. The training trial means the trial from which the MUAPT features were extracted to train the classifiers. The x-axes in A and B indicate the number of trials used for classifier training and the number of motions used for classification, respectively. Twelve motions were used for classification in A, and seven out of eight trials were used to train the classifier in B. The classification results of features are shown with different colors and symbols.

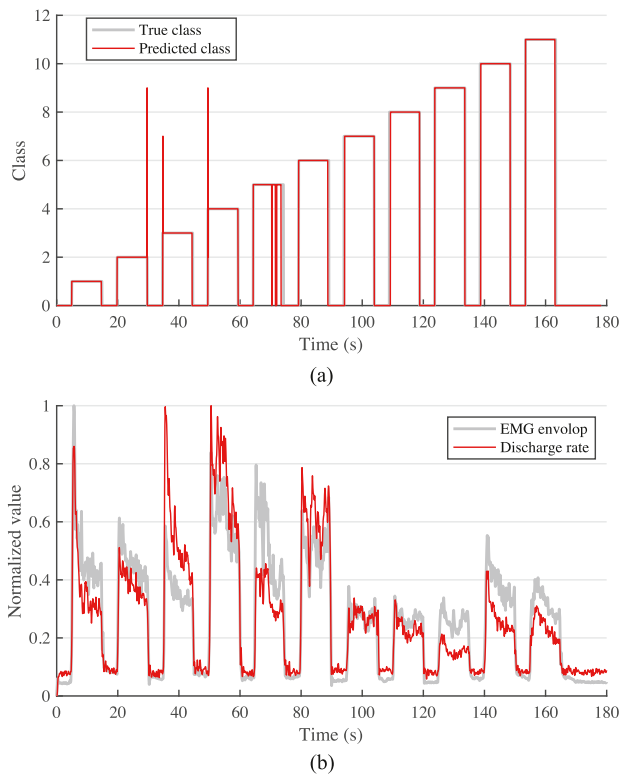

Fig. 10. Offline classification example in a trial. (a) The output of the classifier and the ground truth in a trial where the subject 11 performed twelve motions. Motion 12 (rest) is regarded as class 0. (b) The normalized discharge rate of the CST and the EMG envelope in the same trial as in A.

decoded MUAPTs across trials/tasks. To deal with this issue, a MUAP-based method was proposed to associate the MUs with multiple motor tasks in our previous work [26]. However, the MUAPs are usually extracted by averaging the EMG signals of over 5s [31], prohibiting its online applications due to the computation complexity of the decomposition algorithm. In this work, we improved the decomposition algorithm by pre-estimating the MU filters for each motion. The MU filters were applied directly to the global EMG, allowing to decode and trace the MUs across motor tasks in real-time. Afterward, an online classification method was proposed based on the decomposition results.

The limited number of identified motor units also prohibits the wide application of neural decoding during multiple motor tasks. The blind source separation algorithm usually converges to local solutions towards the large MUAPs in EMG decomposition [35]. The number of identified MUs usually remains relatively stable even though more MUs are activated. Therefore, the decoded neural information appears insufficient during multiple motor tasks. With the motion-wise decomposition method, more MUs with small MUAPs can be identified from local EMG recorded during individual motor tasks, providing a possible solution

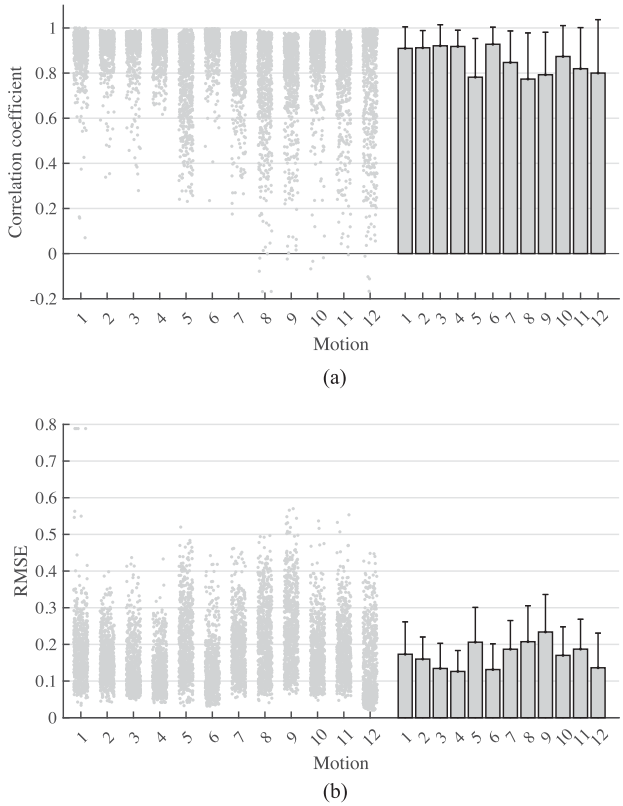

Fig. 11. Results of correlation analysis between the normalized CST discharge rate and EMG envelope for each motion. (a) and (b) give the correlation coefficient and RMSE, respectively. The scatters show the correlation coefficient (a) and RMSE (b) for each motion from all trials, while the bars show the average value with standard deviation.

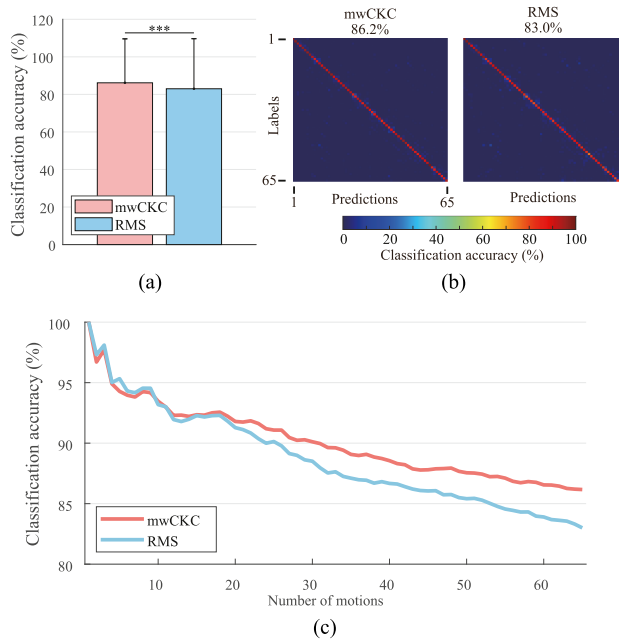

Fig. 12. Classification results of EMG database. (a) and (b) illustrate the average classification accuracy and confusion matrices for 65 gestures, respectively. The variation of classification accuracy along with the number of selected motions is shown in (c).

for this issue. In the CKC-based decomposition method, the critical step is to calculate the MU filter (separation vector) for each MU [36], [37], [38]. The MU filter reflects the correlation between MUAPTs and EMG signals and can be estimated by averaging the EMG signals $y_i(n)$ at all discharge instants of t_j. As the EMG signals are extended, the MU filter also contains the amplitude information around the discharge instants. Therefore, the estimated MU filter can be regarded as the MUAP with ten-sample-point action potential waveforms, reflecting the EMG representation of MU discharges on multiple channels. The EMG representation in a stable mixture model should be time-invariant, and therefore, the MU filter should be approximately constant for EMG decomposition from individual or multiple motor tasks. More MUs with high PNR values were identified, demonstrating the efficiency and superiority of the proposed motion-wise decomposition method. In addition, the MU filters allow for the MU tracking across motor tasks [27], [28], [37]. The MUAPTs were estimated by directly multiplying the pre-trained MU filters with preprocessed EMG signals. The computation load is low enough (<5 ms in a 200-ms window) for the online decomposition. There is sufficient time for the following analysis (feature extraction and classification), providing the possibility of real-time hand gesture recognition based on neural decoding.

Our previous work demonstrated that the decomposition accuracy, whether mwCKC or classic CKC, decreased with the increase in involved MU populations [29]. Compared with the classic CKC, the decrease in the accuracy was severer for the mwCKC, which might result in the low matching rate between the two methods. The motion-wise decomposition was proposed based on the assumption that the mixture procedure of EMG was stationary [29], where the constant MUAP in the stationary mixture guarantees the efficiency of MU filters across motor tasks. Theoretically, the derived classification scheme is limited to isometric conditions. Nonetheless, the experimental protocol in this work also included short-term dynamic contractions, such as the shift between the motions and rest. The results demonstrate a possibility for the motion-wise decomposition to be efficient in this situation, but requires further validation.

B. Classification by Neural Decoding

Different muscle groups are usually involved in multiple motions, resulting in the activation of multiple MU populations [24], [26]. The weighted-averaging MUAP in Fig. 5 illustrates the difference in the activation zone of different MU populations. It should be noted that it is difficult to discriminate all the motor tasks based on the extracted MUAPs. Accordingly, we extracted the neural feature from MUAPTs and used it for hand gesture recognition. As a result, the decomposition performance (e.g., the accuracy and number) of MUAPTs directly affect the classification performance. When multiple motor tasks were involved, the decrease in accuracy was mainly caused by misidentified MU discharges [29]. The discharges of exclusive MUs, which were only activated during one motor task, were sometimes false-identified during other motor tasks.

These misidentified discharges deteriorate the classification accuracy. It is nearly impossible to identify the ground truth in the experiments. Under this situation, the quantity of MUAPTs becomes an important factor.

In this work, the MUs identified by the classic CKC decomposition method were insufficient for gesture recognition, and the corresponding classification accuracy was unacceptable. Actually, the direct application of the CKC to the entire signal is unfair since the EMG signals containing multiple movements do not meet the initial assumption (e.g., stability) of the CKC algorithm. Generally, the CKC should be implemented for each motor task individually. We direct applied the CKC to the entire signal to track motor unit discharges across segments, but could only obtain a few MUAPTs. The previous simulation analysis showed that the classic CKC preferred identifying common MUs which were activated across the entire task. Compared with the classic CKC, the mwCKC could significantly increase the motor unit number at the cost of a decrease in accuracy [29]. Under this situation, the increased MUs contained more information about each gesture. There is a balance between the quantity and quality of the MUAPTs, while the classification results in this study indicate that the mwCKC is more suitable for hand gesture recognition.

The classification accuracy was significantly affected by the size of the training set. However, the average classification accuracy was still higher than 89% when using only one trial for training. Moreover, the classification accuracy could be improved with post-processing methods for the classifier output. For example, a voting machine for the output sequence could be set to filtrate the class outliers shown in Fig. 10(a). Here, we only presented the specific application of the proposed method, while the neural-decoding scheme has the potential to be generalized to other human movements. Compared with the previous study [26], the proposed method exhibited a comparative classification accuracy but allowed for real-time application. As to the 65 gestures, the classification accuracy using the neural feature was lower than that in the previous study [39], where more EMG features were extracted for classifier training. It should be noted that we have not implemented the optimization on the parameter selection of the classifiers. The common parameters such as the linear kernel for SVM and the Euclidean distance for KNN may lead to low classification accuracy, as shown in the supplementary materials. In this work, we focused more on the neural features for gesture recognition rather than the performance of various classifiers.

The classic hand gesture recognition usually used a few electrode pairs and achieved high classification accuracy. However, the statistic features, whether in time- or frequency-domain, are generally the estimation of neural drive and could only partially interpret the neural activities. In this work, we tried to decode the motor unit action potential trains, which directly represent the neural drive transferred from the spinal cord into the muscles. By EMG decomposition, we could observe and analyze the activities of each motor unit. There are some MUs activated in only one motion. These exclusive MUs contain ideal information for gesture recognition [18], [23]. However, few exclusive MUs were identified for most motor tasks, failing to be used for classification. Since a MU would be recruited during more than one task, it becomes nearly impossible to track the exclusive MUs with increased number of motions. In several trials, the subjects might activate a few muscles to maintain the rest postures rather than totally relax. Therefore, a few MUs would be identified during the rest motion. Nevertheless, the number of MUs identified during rest was obviously lower than that of other motions, resulting in a nearly perfect classification accuracy of rest (99.03±4.60%).

C. Limitations

The classification scheme was proposed based on the motion-wise decomposition. The decomposition accuracy would dramatically affect the classification. It should be noted that there is a slight decrease in PNR compared with previous work [26], [30]. Our previous work also indicates that the motion-wise decomposition should be further investigated to improve its reliability [29]. In addition, the decomposition scheme, especially in the online decomposition phase, would be affected by the electrode shift. Therefore, the MU filters and classifier need retraining after replacing the electrode grids. As the classification is based on the motor unit discharges, the long-time tracking technique provides a possible solution [40], which would be investigated in the future work.

Although obtaining promising performance with the EMG decomposition, the placement of such a cumbersome recording system remains challenging. Also, the computation complexity, especially in the decomposition of the offline training phase, is much higher than the classic methods. In this work, the performance of the proposed gesture recognition scheme was only evaluated in offline analysis and virtual environment. The high classification accuracy does not strictly indicate good performance in practical applications. Therefore, the proposed method requires further validation in daily tasks. Currently, one of the most significant challenges in myoelectric control is to provide simultaneous and proportional control (SPC), where both the motor class and the corresponding activation level should be recognized. As the smallest functional unit in human movements, the MU discharges contain the neural drive transferred into the muscles. The activation of the MU pool reflects the class of the activated motions [18], and the discharge rate of the CST could reflect the activation level [41]. The high correlation between the CST discharge rate and EMG envelope indicated the feasibility of SPC based on EMG decomposition. The performance of SPC based on neural decoding will be investigated in future work.

V. Conclusion

We have proposed a real-time hand gesture recognition method by decoding MU discharges from high-density surface EMG signals in a motion-wise way. On average, 164±34 MUs were identified for twelve hand gestures from each subject. The time delay of EMG decomposition was less than 5 ms, allowing for real-time applications. Moreover, the neural features extracted from motor unit activities exhibited a classification accuracy of up to 94.6% for 12 gestures and 86.2% for 65

gestures. These results support the feasibility and superiority of the proposed method in real-time MU identification and gesture recognition across multiple motor tasks (>10), promoting the broad application of neural decoding for human-machine interfacing.

Appendix
EMG Decomposition Algorithm

A motion-wise decomposition scheme based on the convolution kernel compensation (CKC) was used to identify MU discharges across motor tasks and trials [15], [36]. In the CKC-based decomposition, the EMG signals are modeled as the multi-input-multi-output system as:

$$y_i(n) = \sum_{j=1}^{J}\sum_{l=0}^{L-1} h_{ij}(l)t_j(n-l) + \omega_i(n) \quad (1)$$
$$i = 1, 2, \ldots, M$$

where $y_i(n)$ is the ith EMG channel, n is the discrete time as sample point, $h_{ij}(l)$ is the action potential of the jth MU recorded at the channel i, $t_j(n)$ is the spike train of the jth MU (the jth MUAPT), $\omega_i(n)$ is the additive noise at channel i, L is the sample length of the action potentials, and J and M is the number of active MUs and EMG channels, respectively. The EMG signals are usually extended by n delayed repetitions to increase the ratio between the number of observations and sources [42]. $n = 10$ was selected in this work.

The CKC algorithm compensates the unknown mixture kernel ($h_{ij}(l)$) and estimates the MUAPT as:

$$\widehat{t}_j(n) = f_{t_j y}^T C_{yy}^{-1} y(n) \quad (2)$$

where $C_{yy} = E(y(n)y^T(n))$ is the correlation matrix of EMG signals, $f_{t_j y} = E(y(n)t_j^T(n))$ is the cross-correlation vector between EMG signals and the jth spike train, and $E(\cdot)$ denotes mathematical expectation. The cross-correlation vector ($f_{t_j y}$) can be regarded as a MU filter for processed EMG signals ($C_{yy}^{-1}y(n)$) to estimate the discharge train. Assuming the mixing process is ergodic, the MU filter can be estimated by averaging the EMG signals ($y_i(n)$) at all discharge instants of t_j [15].

In the motion-wise decomposition algorithm, the EMG signals are divided into multiple segments depending on the activation duration of each motor task ($y = [y_1, y_2, \ldots]$, y_i indicates EMG signals from the ith motion). The CKC is implemented individually for each segment. In this work, the MU filter is first calculated iteratively using the natural gradient descent method in each segment [36]. Then the K-means clustering was used to extract the discharges from the estimated MUAPTs. A silhouette measure (SIL) was computed to evaluate the extraction quality. The SIL was defined as the difference between the within-cluster sums of point-to-centroid distances and the same measure calculated between clusters [42]. Only MUAPTs with SIL > 0.8 were kept for the following analysis. A signal-based metric, pulse-to-noise ratio (PNR), was applied to evaluate the decomposition accuracy of experimental EMG signals [30].

$$\mathrm{PNR}_j = 10 \cdot \log\left(\frac{E\left(\widehat{t}_j(n)|_{\widehat{t}_j(n)\geq r}\right)}{E\left(\widehat{t}_j(n)|_{\widehat{t}_j(n)<r}\right)}\right) \quad (3)$$

where $E\left(x|_{\widehat{t}_j(n)\geq r}\right)$ and $E\left(x|_{\widehat{t}_j(n)<r}\right)$ denote the mean across all time moments in which the jth MU is estimated to have or have not discharged, respectively [30]. r indicates the threshold to extract discharges from estimated MUAPTs.

The MU filters from each segment are stored and used to decompose the global EMG data depending on (2), while the correlation matrix (C_{yy}) is calculated using the global EMG signals. The MU filters of MUs activated during multiple motions would be identified more than once, resulting in the MUAPT duplications from the global EMG. Among the duplications, the MUAPT with a lower PNR would be discarded. Two MUAPTs were regarded as duplicated if the RoA was higher than 0.3 [43]. The RoA was calculated as $RoA = \frac{n_0}{n_0+n_1+n_2}$, where n_0 indicates the number of concurrent discharges (tolerance in discharge timing ± 0.5 ms), n_1 and n_2 are numbers of identified discharges in the two MUAPTs, respectively. The alignment in a range of ± 50 ms was required before matching two MUAPTs. The decomposition parameters (e.g., MU filters, correlation matrix, and clustering centroids) of duplicated MUAPTs were also discarded. There were usually $< 30\%$ MUAPTs removed in this step.

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